

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Mixed

03

10 answers · Rationale-rich · Teach from doc

Q1**30**

The correct answer is 30. The situation can be represented by the equation $x(2^4) = 480$, where the 2 represents the fact that the amount of money in the account doubled each year and the 4 represents the fact that there are 4 years between January 1, 2001, and January 1, 2005. Simplifying $x(2^4) = 480$ gives $16x = 480$. Therefore, $x = 30$.

Q2**.54**

Accept also: 27/50

The correct answer is .54 . It's given that the number a is 70% less than the positive number b . Therefore, $a = 1 - \frac{70}{100}b$, which is equivalent to $a = 1 - 0.70b$, or $a = 0.30b$. It's also given that the number c is 80% greater than a . Therefore, $c = 1 + \frac{80}{100}a$, which is equivalent to $c = 1 + 0.80a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.80a$ yields $c = 1.80(0.30b)$, or $c = 0.54b$. Thus, c is 0.54 times b . Note that .54 and 27/50 are examples of ways to enter a correct answer.

Q3

C

C. $f(t) = -\frac{21}{130}t + 4$

Choice C is correct. It is assumed that the oil and gas production decreased at a constant rate. Therefore, the function f that best models the production t years after the year 2000 can be written as a linear function, $f(t) = mt + b$, where m is the rate of change of the oil and gas production and b is the oil and gas production, in millions of barrels, in the year 2000. Since there were 4 million barrels of oil and gas produced in 2000, $b = 4$. The rate of change, m , can be calculated as $\frac{4 - 1.9}{0 - 13} = -\frac{2.1}{13}$, which is equivalent to $-\frac{21}{130}$, the rate of change in choice C.

Choices A and B are incorrect because each of these functions has a positive rate of change. Since the oil and gas production decreased over time, the rate of change must be negative. Choice D is incorrect. This model may result from misinterpreting 1.9 million barrels as the amount by which the production decreased.

Q4

C

C.

x	y
3	16
5	28
7	40

Choice C is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 3 for x in the given inequality yields $y < 63 + 2$, or $y < 20$. Therefore, when $x = 3$, the corresponding value of y is less than 20. Substituting 5 for x in the given inequality yields $y < 65 + 2$, or $y < 32$. Therefore, when $x = 5$, the corresponding value of y is less than 32. Substituting 7 for x in the given inequality yields $y < 67 + 2$, or $y < 44$. Therefore, when $x = 7$, the corresponding value of y is less than 44. For the table in choice C, when $x = 3$, the corresponding value of y is 16, which is less than 20; when $x = 5$, the corresponding value of y is 28, which is less than 32; when $x = 7$, the corresponding value of y is 40, which is less than 44. Therefore, the table in choice C gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is 20, which is not less than 20; when $x = 5$, the corresponding value of y is 32, which is not less than 32; when $x = 7$, the corresponding value of y is 44, which is not less than 44.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is 36, which is not less than 32.

Choice D is incorrect. In the table for choice D, when $x = 3$, the corresponding value of y is 24, which is not less than 20; when $x = 5$, the corresponding value of y is 36, which is not less than 32; when $x = 7$, the corresponding value of y is 48, which is not less than 44.

Q5

D

D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$
4	$\frac{92}{15}$

Choice D is correct. Each of the tables gives the same three values of x : 1, 2, and 4. Substituting 1 for x in the given equation yields $\frac{3}{5} \cdot 1 + \frac{3}{4}y = 7$, or $\frac{3}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{3}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{32}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{128}{15}$.

Therefore, when $x = 1$, the corresponding value of y for the given equation is $\frac{128}{15}$. Substituting 2 for x in the given equation yields $\frac{3}{5} \cdot 2 + \frac{3}{4}y = 7$, or $\frac{6}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{6}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{29}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{116}{15}$.

Therefore, when $x = 2$, the corresponding value of y for the given equation is $\frac{116}{15}$. Substituting 4 for x in the given equation yields $\frac{3}{5} \cdot 4 + \frac{3}{4}y = 7$, or $\frac{12}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{12}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{23}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{92}{15}$.

Therefore, when $x = 4$, the corresponding value of y for the given equation is $\frac{92}{15}$. The table in choice D gives x -values of 1, 2, and 4 and corresponding y -values of $\frac{128}{15}$, $\frac{116}{15}$, and $\frac{92}{15}$, respectively.

Therefore, the table in choice D gives three values of x and their corresponding values of y for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4} + y = 7$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + y = 10$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4}y = 8$.

Q6**.8823**

Accept also: .8824, 15/17

The correct answer is $\frac{15}{17}$. It's given that angle J is the right angle in triangle JKL . Therefore, the acute angles of triangle JKL are angle K and angle L . The hypotenuse of a right triangle is the side opposite its right angle. Therefore, the hypotenuse of triangle JKL is side KL . The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to the angle to the length of the hypotenuse. It's given that $\cos K = \frac{24}{51}$. This can be written as $\cos K = \frac{8}{17}$. Since the cosine of angle K is a ratio, it follows that the length of the side adjacent to angle K is $8n$ and the length of the hypotenuse is $17n$, where n is a constant. Therefore, $JK = 8n$ and $KL = 17n$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. For triangle JKL , it follows that $JK^2 + JL^2 = KL^2$. Substituting $8n$ for JK and $17n$ for KL yields $8n^2 + JL^2 = 17n^2$. This is equivalent to $64n^2 + JL^2 = 289n^2$. Subtracting $64n^2$ from each side of this equation yields $JL^2 = 225n^2$. Taking the square root of each side of this equation yields $JL = 15n$. Since $\cos L = \frac{JL}{KL}$, it follows that $\cos L = \frac{15n}{17n}$, which can be rewritten as $\cos L = \frac{15}{17}$. Note that $15/17$, $.8824$, $.8823$, and 0.882 are examples of ways to enter a correct answer.

Q7**D****D. 1,944**

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle ABCD, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of ABCD can be represented by the expression xy . It's given that the length of each side of EFGH is 6 times the length of the corresponding side of ABCD. Therefore, the length, in units, of the base of EFGH can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $6x6y$, which is equivalent to $36xy$. Therefore, the area, in square units, of EFGH can be represented by the expression $36xy$. It's given that the area of ABCD is 54 square units. Since xy represents the area, in square units, of ABCD, substituting 54 for xy in the expression $36xy$ yields 3654, or 1,944. Therefore, the area, in square units, of EFGH is 1,944.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not 6, times the length of the corresponding side of ABCD.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not 6, times the length of the corresponding side of ABCD.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not 6, times the length of the corresponding side of ABCD.

Q8**C****C. 0**

Choice C is correct. Since the expression $x^2 + 6x - 7$ can be factored as $(x + 7)(x - 1)$, the given equation can be rewritten as $\frac{(x + 7)(x - 1)}{x + 7} = ax + d$. Since $x \neq -7$, $x + 7$ is also not equal to 0, so both the numerator and denominator of $\frac{(x + 7)(x - 1)}{x + 7}$ can be divided by $x + 7$. This gives $x - 1 = ax + d$. Equating the coefficient of x on each side of the equation gives $a = 1$. Equating the constant terms gives $d = -1$. The sum is $1 + (-1) = 0$.

Choice A is incorrect and may result from incorrectly simplifying the equation. Choices B and D are incorrect. They are the values of d and a , respectively, not $a + d$.

Q9**7**

The correct answer is 7. When an equation is of the form $y = ax^2 + bx + c$, where a , b , and c are constants, the value of y reaches its minimum when $x = -\frac{b}{2a}$. Since the given equation is of the form $y = ax^2 + bx + c$, it follows that $a = 1$, $b = -14$, and $c = 22$. Therefore, the value of y reaches its minimum when $x = -\frac{-14}{2(1)}$, or $x = 7$.

Q10

D

D. Neither I nor II

Choice D is correct. It's given that the graph of $y = fx$ in the xy -plane is a parabola with vertex h, k . If $f(-9) = f(3)$, then for the graph of $y = fx$, the point with an x -coordinate of -9 and the point with an x -coordinate of 3 have the same y -coordinate. In the xy -plane, a parabola is a symmetric graph such that when two points have the same y -coordinate, these points are equidistant from the vertex, and the x -coordinate of the vertex is halfway between the x -coordinates of these two points. Therefore, for the graph of $y = fx$, the points with x -coordinates -9 and 3 are equidistant from the vertex, h, k , and h is halfway between -9 and 3 . The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3 . Therefore, $h = -3$. The equation defining f can also be written in vertex form, $fx = ax - h^2 + k$. Substituting -3 for h in this equation yields $fx = ax - (-3)^2 + k$, or $fx = ax - 9 + k$. This equation is equivalent to $fx = ax^2 + 6x + 9 + k$, or $fx = ax^2 + 6ax + 9a + k$. Since $fx = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9(\frac{2}{3}) + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c - 6$. If $k < 0$, then $c - 6 < 0$, or $c < 6$. Since c could be any value less than 6 , it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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